

Exam Study Guide — Multiple Choice

Dark Web Overview & Military Cyber Defense Tools — 60 Q&A with Answer Key

SECTION 1 — Clear Web, Deep Web & Dark Web: Definitions

1. What are the three layers of the internet, from most to least publicly accessible?

- A) Clear Web, Deep Web, Dark Web
- B) Deep Web, Dark Web, Clear Web
- C) Dark Web, Clear Web, Deep Web
- D) Surface Web, Dark Web, Private Web

Answer: A) Clear Web, Deep Web, Dark Web

Dark Web $\subset$ Deep Web $\subset$ Internet — Clear Web is indexed/open, Deep Web is unindexed but reachable, Dark Web is intentionally hidden.

2. What defines the Clear Web?

- A) Content publicly indexable by search engines, accessible via regular browsers
- B) Content hidden behind Tor and .onion addresses
- C) Content requiring authentication like banking portals
- D) Content only accessible via encrypted messaging apps

Answer: A) Content publicly indexable by search engines, accessible via regular browsers

3. What defines the Deep Web?

- A) Publicly indexed news and social media
- B) Content not indexed by search engines but reachable via normal browsers (e.g., banking, email)
- C) Content requiring Tor to access
- D) Illegal marketplaces only

Answer: B) Content not indexed by search engines but reachable via normal browsers (e.g., banking, email)

The Deep Web is the vast majority of online content by volume.

4. What defines the Dark Web?

- A) Any website using HTTPS
- B) The portion of the internet blocked by governments
- C) An intentionally hidden subset of the Deep Web requiring special software like Tor and .onion addressing
- D) Content indexed by Google but paywalled

Answer: C) An intentionally hidden subset of the Deep Web requiring special software like Tor and .onion addressing

SECTION 2 — History & Onion Routing (Tor)

5. Who originally developed onion routing, and why?

- A) The Tor Project non-profit, for commercial privacy products
- B) The U.S. Naval Research Laboratory, to protect government communications
- C) A group of hackers, to sell drugs anonymously
- D) Facebook, to protect user data

Answer: B) The U.S. Naval Research Laboratory, to protect government communications

6. What happened to Tor in the 2000s?

- A) It was banned worldwide
- B) The Tor Project became a non-profit and released software/hidden services publicly
- C) It was sold to a private company
- D) It was merged with the Deep Web

Answer: B) The Tor Project became a non-profit and released software/hidden services publicly

7. What is the basic mechanism of onion routing?

- A) Traffic bounces randomly with no encryption
- B) Traffic is encrypted once and sent directly to the destination
- C) Traffic is encrypted in multiple layers and routed through relays, each peeling one layer
- D) Traffic is compressed and cached at each node

Answer: C) Traffic is encrypted in multiple layers and routed through relays, each peeling one layer

8. Why can no single relay node see both the origin and destination of Tor traffic?

- A) Nodes are randomly shut down
- B) Each node decrypts only its own layer, learning only the previous/next hop
- C) All relays are government-controlled
- D) The destination server hides its own IP

Answer: B) Each node decrypts only its own layer, learning only the previous/next hop

SECTION 3 — Purpose, Legitimate & Illicit Users

9. Which of the following is a legitimate use of Dark Web / anonymity technology?

- A) Operating an illegal drug marketplace
- B) Journalists protecting sources via SecureDrop
- C) Distributing malware
- D) Money laundering via cryptocurrency mixers

Answer: B) Journalists protecting sources via SecureDrop

10. What tool is specifically named for enabling secure journalist-source communication?

- A) SecureDrop
- B) Tor Browser Bundle alone
- C) A Bitcoin wallet
- D) A generic VPN app

Answer: A) SecureDrop

11. How do citizens in authoritarian regimes benefit from Dark Web / Tor technology?

- A) They can buy weapons at reduced risk
- B) They gain access to uncensored news, education, and communication channels
- C) They can trade stolen credentials safely
- D) They can avoid taxes

Answer: B) They gain access to uncensored news, education, and communication channels

12. Which of the following is an illicit use of the Dark Web?

- A) Threat intelligence monitoring
- B) Coordination of ransomware-as-a-service operations
- C) Academic study of markets and anonymity
- D) Human rights documentation

Answer: B) Coordination of ransomware-as-a-service operations

SECTION 4 — Security, Anonymity & Risks

13. Which of the following is a way onion routing protects users?

- A) It guarantees no malware can ever reach the user
- B) Multi-layer encryption hides the path between user and destination from most observers
- C) It automatically verifies every website's legitimacy
- D) It blocks all law enforcement monitoring permanently

Answer: B) Multi-layer encryption hides the path between user and destination from most observers

14. What is the key weakness of the exit node in a Tor circuit?

- A) It cannot connect to the open internet
- B) It can observe unencrypted traffic leaving the network if the destination isn't using HTTPS
- C) It automatically logs the user's real IP address
- D) It encrypts traffic twice, causing errors

Answer: B) It can observe unencrypted traffic leaving the network if the destination isn't using HTTPS

15. What is the most common real-world cause of users being de-anonymized on the Dark Web?

- A) Tor's encryption being mathematically broken
- B) Operational security mistakes (reused IDs, personal info, malware) combined with investigative techniques
- C) Random server failures
- D) Government-mandated backdoors in all browsers

Answer: B) Operational security mistakes (reused IDs, personal info, malware) combined with investigative techniques

16. Which of the following is a risk associated with using the Dark Web?

- A) Guaranteed anonymity with no possible legal consequences
- B) Illegal content exposure, scams, malware, and legal exposure even from simply visiting certain sites
- C) Automatic antivirus protection
- D) Verified vendor identities on all marketplaces

Answer: B) Illegal content exposure, scams, malware, and legal exposure even from simply visiting certain sites

SECTION 5 — Marketplaces, Future Outlook & Professional Implications

17. Which of the following is a typical structural feature of anonymous Dark Web marketplaces?

- A) Government-issued vendor licenses
- B) Escrow systems holding funds until delivery is confirmed
- C) Publicly verifiable real names for all vendors
- D) Fixed prices set by international regulators

Answer: B) Escrow systems holding funds until delivery is confirmed

18. Why did marketplaces shift from Bitcoin toward privacy-focused cryptocurrencies over time?

- A) Bitcoin transactions became too expensive
- B) To reduce traceability of financial transactions on the public blockchain
- C) Privacy coins are legally required
- D) Bitcoin was banned globally

Answer: B) To reduce traceability of financial transactions on the public blockchain

19. According to the 'Future Outlook,' what is the expected long-term dynamic between anonymity tools and law enforcement?

- A) Anonymity tools will inevitably disappear
- B) An ongoing 'arms race' where advances in one prompt responses in the other
- C) Law enforcement will stop investigating cybercrime
- D) Anonymity technology will fully merge with mainstream browsers within a year

Answer: B) An ongoing 'arms race' where advances in one prompt responses in the other

20. What is the 'dual-use dilemma' as it relates to anonymity technology?

- A) The tools cost twice as much as regular software
- B) The same tools that protect journalists and activists also shield criminals, complicating broad restriction
- C) Users must choose between Tor and I2P
- D) All anonymity tools require two forms of ID

Answer: B) The same tools that protect journalists and activists also shield criminals, complicating broad restriction

SECTION 6 — Cyber Defense as Mission Assurance

21. According to the military cyber defense material, cyber defense is best described as what kind of function?

- A) A purely administrative IT support function
- B) A warfighting capability ensuring forces can execute missions in contested cyberspace
- C) An optional add-on with no operational relevance
- D) A function relevant only during peacetime training

Answer: B) A warfighting capability ensuring forces can execute missions in contested cyberspace

22. Define 'Mission Assurance.'

- A) A process to protect or ensure continued function/resilience of capabilities and assets critical to DoD mission-essential functions
- B) A guarantee that no cyberattack will ever succeed
- C) A checklist for antivirus installation
- D) A financial audit process for IT budgets

Answer: A) A process to protect or ensure continued function/resilience of capabilities and assets critical to DoD mission-essential functions

23. In tactical environments, which CIA principles often matter more than pure secrecy?

- A) Confidentiality alone
- B) Availability and integrity
- C) Neither — secrecy is always most important
- D) Authenticity alone

Answer: B) Availability and integrity

24. How is 'success' measured under Mission Assurance, per the Measurable Outcomes concept?

- A) The number of blocked packets
- B) Mission completion under cyber attack
- C) Zero intrusions detected
- D) Total number of patched vulnerabilities

Answer: B) Mission completion under cyber attack

SECTION 7 — Core Principles: CIA + Extensions

25. What does the 'C' in the CIA triad stand for, and what does it protect against in a military context?

- A) Confidentiality — preventing unauthorized disclosure, critical for OPSEC/classified data
- B) Continuity — ensuring uptime
- C) Compliance — meeting regulations
- D) Command — controlling access

Answer: A) Confidentiality — preventing unauthorized disclosure, critical for OPSEC/classified data

26. What does 'Integrity' ensure, and why can false data be worse than denial?

- A) It ensures fast data transmission; speed matters more than accuracy
- B) It ensures data/systems aren't altered; false data can mislead decision-makers without them realizing it, unlike obvious denial
- C) It ensures encryption keys rotate daily
- D) It ensures backups are stored offsite

Answer: B) It ensures data/systems aren't altered; false data can mislead decision-makers without them realizing it, unlike obvious denial

27. What does 'Availability' mean, and what is a primary wartime cyber effect that targets it?

- A) Authorized access when needed; Denial-of-Service (DoS) is a primary wartime effect against it
- B) Keeping data secret; malware is the primary effect
- C) Verifying identities; phishing is the primary effect
- D) Logging all actions; ransomware is the primary effect

Answer: A) Authorized access when needed; Denial-of-Service (DoS) is a primary wartime effect against it

28. What are the three military extensions added to the CIA triad?

- A) Authenticity, Accountability, Resilience
- B) Encryption, Compression, Redundancy
- C) Firewalls, Antivirus, Patching
- D) Confidentiality, Compliance, Continuity

Answer: A) Authenticity, Accountability, Resilience

29. Define 'Accountability' and explain why it matters legally.

- A) It hides user identities to protect privacy
- B) It attributes actions to specific actors via logging/audit trails, enabling non-repudiation, forensics, and legal process
- C) It ensures systems never go offline
- D) It verifies hardware authenticity only

Answer: B) It attributes actions to specific actors via logging/audit trails, enabling non-repudiation, forensics, and legal process

SECTION 8 — Defense-in-Depth

30. What is the core idea behind Defense-in-Depth?

- A) Relying on one very strong firewall
- B) Layered security so failure of any single control doesn't cause total compromise
- C) Outsourcing all security to a single vendor
- D) Physical security alone is sufficient

Answer: B) Layered security so failure of any single control doesn't cause total compromise

31. Which of these correctly pairs a Defense-in-Depth layer with an example control?

- A) Physical & Environmental — Facility access control, TEMPEST shielding
- B) Identity & Access — Firewalls and DMZs
- C) Network Perimeter — EDR and application allow-listing
- D) People & Process — Encryption and secure coding

Answer: A) Physical & Environmental — Facility access control, TEMPEST shielding

32. Give an example control for the 'Host & Endpoint' layer.

- A) Firewalls and boundary defense
- B) Endpoint Detection and Response (EDR) and application allow-listing
- C) Facility access control
- D) Security awareness training

Answer: B) Endpoint Detection and Response (EDR) and application allow-listing

33. Give an example control for the 'Identity & Access' layer.

- A) Authentication, authorization, least privilege
- B) TEMPEST shielding
- C) Secure coding practices
- D) Firewalls

Answer: A) Authentication, authorization, least privilege

34. Why is relying on only one security control dangerous?

- A) It's more expensive than multiple controls
- B) If that one control fails or is bypassed, nothing stops total compromise of the mission system
- C) Single controls are illegal under Title 10
- D) It slows down network performance

Answer: B) If that one control fails or is bypassed, nothing stops total compromise of the mission system

SECTION 9 — Zero Trust Architecture

35. What is the core motto of Zero Trust architecture?

- A) "Trust but verify occasionally"
- B) "Never trust, always verify. Assume breach."
- C) "Trust internal networks, verify external only"
- D) "Verify once at login, trust thereafter"

Answer: B) "Never trust, always verify. Assume breach."

36. What does 'Explicit Verification' mean under Zero Trust?

- A) Authenticate/authorize using all available data — identity, device health, location, behavior
- B) Only check the username and password
- C) Verify identity once per year
- D) Trust any device on the corporate network automatically

Answer: A) Authenticate/authorize using all available data — identity, device health, location, behavior

37. What does 'Least Privilege Access' mean, and what two access models support it?

- A) Granting broad access to speed up work; supported by SSO and MFA
- B) Granting only minimum access needed; supported by Just-In-Time (JIT) and Just-Enough-Access (JEA)
- C) Granting admin rights to all users; supported by RBAC
- D) Granting access based on seniority only

Answer: B) Granting only minimum access needed; supported by Just-In-Time (JIT) and Just-Enough-Access (JEA)

38. What does 'Assume Breach' mean in practice?

- A) Assume no attacker will ever succeed
- B) Minimize blast radius via network segmentation, end-to-end encryption, and analytics-based detection
- C) Disable all monitoring to reduce noise
- D) Trust all internal traffic since breach is impossible

Answer: B) Minimize blast radius via network segmentation, end-to-end encryption, and analytics-based detection

39. Scenario: A legitimate employee's password is correct, but they're logging in from an unfamiliar device and accessing unusual files. What should happen under Zero Trust?

- A) Automatically grant full access since the password was correct
- B) Factor in device health, location, and behavior, and apply extra scrutiny given the anomaly
- C) Permanently lock the account with no review
- D) Ignore the anomaly since it's a known employee

Answer: B) Factor in device health, location, and behavior, and apply extra scrutiny given the anomaly

SECTION 10 — Continuous Monitoring

40. What is the purpose of Continuous Monitoring?

- A) To monitor employees' personal social media
- B) To maintain persistent visibility into security posture and detect deviations from baseline in near real time
- C) To replace all human analysts with automation
- D) To generate compliance paperwork only

Answer: B) To maintain persistent visibility into security posture and detect deviations from baseline in near real time

41. What does 'Telemetry Collection' involve?

- A) Fusing logs, network flows, endpoint sensors, vulnerability data, and threat intel into one picture
- B) Only collecting firewall logs
- C) Manually reviewing paper reports
- D) Encrypting all outbound traffic

Answer: A) Fusing logs, network flows, endpoint sensors, vulnerability data, and threat intel into one picture

42. Why is risk-based scoring important for 'Alerting & Priority'?

- A) It focuses analysts' attention on threats to mission-critical assets rather than treating all alerts equally
- B) It eliminates the need for any human review
- C) It automatically blocks all traffic
- D) It reduces the number of sensors needed

Answer: A) It focuses analysts' attention on threats to mission-critical assets rather than treating all alerts equally

43. What is the key message of the 'Human-Machine Team' concept?

- A) Tools should fully replace human analysts
- B) Tools amplify operators; they do not replace judgment, context, or escalation
- C) Humans should ignore tool outputs entirely
- D) Machines are only useful for reporting, not detection

Answer: B) Tools amplify operators; they do not replace judgment, context, or escalation

SECTION 11 — Detection-to-Response Cycle

44. List the four stages of the Detection-to-Response cycle in order.

- A) Detect, Analyze, Respond, Adapt
- B) Analyze, Detect, Adapt, Respond
- C) Respond, Detect, Analyze, Adapt
- D) Adapt, Respond, Detect, Analyze

Answer: A) Detect, Analyze, Respond, Adapt

45. What happens during the 'Analyze' stage?

- A) Sensors first flag potential malicious activity
- B) Triage and correlation to determine scope, severity, and mission impact
- C) Malware removal and vulnerability patching
- D) Updating playbooks after the fact

Answer: B) Triage and correlation to determine scope, severity, and mission impact

46. What happens during the 'Respond' stage?

- A) Sharing lessons learned with partners
- B) Containment, eradication, and recovery — applying tactical/operational countermeasures
- C) Initial sensor-based detection only
- D) Long-term policy development

Answer: B) Containment, eradication, and recovery — applying tactical/operational countermeasures

47. What determines whether an intrusion becomes a 'mission-impacting event'?

- A) The number of analysts on shift
- B) The speed and accuracy of the Detection-to-Response cycle
- C) Whether the attacker is a nation-state actor
- D) The time of day the attack occurs

Answer: B) The speed and accuracy of the Detection-to-Response cycle

SECTION 12 — Response Actions in Detail

48. What actions occur during 'Containment'?

- A) Restore from backups and validate integrity
- B) Isolate hosts/networks, block C2, revoke credentials, limit lateral movement while preserving forensic evidence
- C) Conduct after-action review
- D) Update long-term security policy

Answer: B) Isolate hosts/networks, block C2, revoke credentials, limit lateral movement while preserving forensic evidence

49. What actions occur during 'Eradication'?

- A) Remove malware, close vulnerabilities, reset compromised accounts, eliminate persistence mechanisms
- B) Isolate affected hosts only
- C) Inform command authority
- D) Share threat indicators with allies

Answer: A) Remove malware, close vulnerabilities, reset compromised accounts, eliminate persistence mechanisms

50. What actions occur during 'Recovery'?

- A) Block command-and-control channels
- B) Restore from known-good backups, validate integrity, return mission capabilities in priority order
- C) Conduct legal review of ROE
- D) Draft new cyber policy frameworks

Answer: B) Restore from known-good backups, validate integrity, return mission capabilities in priority order

51. Scenario: Isolating a compromised server would stop an attack but would disrupt a mission-critical function. What must the team weigh before acting, and who should be informed?

- A) Nothing — isolate immediately without informing anyone
- B) Weigh mission impact of isolation vs. continued compromise, and inform command authority before the mission-impacting decision
- C) Only the IT helpdesk needs to know
- D) Wait indefinitely until the attacker leaves voluntarily

Answer: B) Weigh mission impact of isolation vs. continued compromise, and inform command authority before the mission-impacting decision

SECTION 13 — Government & Military Context

52. What do Title 10 and Title 50 define in relation to cyber operations?

- A) Encryption standards for classified data
- B) Who may conduct cyber operations and under what conditions
- C) Budget allocations for cyber commands
- D) Technical specifications for firewalls

Answer: B) Who may conduct cyber operations and under what conditions

53. Name a policy framework that sets ROE and risk management rules for military cyber defense.

- A) DoD Cyber Strategy / RMF (Risk Management Framework)
- B) OSHA workplace safety guidelines
- C) GDPR
- D) PCI-DSS

Answer: A) DoD Cyber Strategy / RMF (Risk Management Framework)

54. Why is interagency and allied coordination (DHS, FBI, IC, partners) important in cyber defense?

- A) It's legally optional and rarely useful
- B) It's essential for shared situational awareness and collective defense across organizational/national boundaries
- C) It only matters for financial audits
- D) It replaces the need for military command authority

Answer: B) It's essential for shared situational awareness and collective defense across organizational/national boundaries

SECTION 14 — Command Authority in Cyber Defense

55. What is 'Unity of Command' in cyber defense?

- A) Every unit acts independently without coordination
- B) Clear lines of authority ensuring defensive cyber ops support the joint force commander without conflicting actions
- C) Only civilian agencies have authority over cyber operations
- D) Command rotates randomly among units

Answer: B) Clear lines of authority ensuring defensive cyber ops support the joint force commander without conflicting actions

56. What is 'Mission Command' as applied to cyber defenders?

- A) Empowering subordinate defenders with commander's intent so they can act rapidly within authorized parameters
- B) Requiring approval from the highest headquarters for every action
- C) Removing all decision-making from local defenders
- D) Delaying all actions until after-action review

Answer: A) Empowering subordinate defenders with commander's intent so they can act rapidly within authorized parameters

57. Who is responsible for accepting residual cyber risk to a mission — the CISO or the commander?

- A) The CISO exclusively
- B) The commander — commanders own residual cyber risk to their missions
- C) Neither — risk is automatically accepted by the system
- D) A third-party contractor

Answer: B) The commander — commanders own residual cyber risk to their missions

SECTION 15 — Critical Thinking & Scenario Synthesis

58. Critical Thinking: 'A cyber defense operation is successful when the attacker is stopped.' Do you agree or disagree, using Mission Assurance?

- A) Agree — stopping the attacker is the only measure of success
- B) Disagree — success should be measured by mission completion under attack, not merely stopping the attacker
- C) Agree — the number of blocked packets is what matters
- D) Disagree — because attackers can never actually be stopped

Answer: B) Disagree — success should be measured by mission completion under attack, not merely stopping the attacker

59. Reflection: What is the difference between protecting a computer network and protecting a mission?

- A) There is no difference — they are the same thing
- B) Protecting a network focuses on technical infrastructure; protecting a mission focuses on the operational objective being achievable despite degradation
- C) Protecting a mission only applies to physical equipment
- D) Protecting a network is more important than protecting a mission

Answer: B) Protecting a network focuses on technical infrastructure; protecting a mission focuses on the operational objective being achievable despite degradation

60. Scenario: The cybersecurity team detects unauthorized access, unusual traffic, and signs of lateral movement toward mission-critical systems, all while a mission is in progress. What overall approach should guide their response?

- A) Immediately shut down the entire network without regard to mission impact
- B) Apply the full Detect→Analyze→Respond→Adapt cycle, prioritize mission-essential systems, weigh mission impact before disruptive actions, keep command authority informed, and apply defense-in-depth/Zero Trust principles throughout
- C) Ignore the mission and focus solely on network cleanliness
- D) Wait for the next scheduled maintenance window to respond

Answer: B) Apply the full Detect→Analyze→Respond→Adapt cycle, prioritize mission-essential systems, weigh mission impact before disruptive actions, keep command authority informed, and apply defense-in-depth/Zero Trust principles throughout

Quick Answer Key

For fast self-check — full answers with explanations are above.

1-A 2-A 3-B 4-C 5-B 6-B 7-C 8-B 9-B 10-A
11-B 12-B 13-B 14-B 15-B 16-B 17-B 18-B 19-B 20-B
21-B 22-A 23-B 24-B 25-A 26-B 27-A 28-A 29-B 30-B
31-A 32-B 33-A 34-B 35-B 36-A 37-B 38-B 39-B 40-B
41-A 42-A 43-B 44-A 45-B 46-B 47-B 48-B 49-A 50-B
51-B 52-B 53-A 54-B 55-B 56-A 57-B 58-B 59-B 60-B